A version of Young's experimental setup—known as the double-slit experiment—is schematically illustrated in Figure 4.3. Feynman was fond of saying that all of quantum mechanics can be gleaned from carefully thinking through the implications of this single experiment, so it's well worth discussing. As we see from Figure 4.3, light is shone on a thin solid barrier in which two slits are cut. A photographic plate records the light that gets through the slits—brighter areas of the photograph indicate more incident light. The experiment consists of comparing the images on photographic plates that result when either or both of the slits in the barrier are kept open and the light source is turned on.

If the left slit is covered and the right slit is open, the photograph looks like that shown in Figure 4.4. This makes good sense, since the light that hits the photographic plate must pass through the only open slit and will therefore be concentrated around the right part of the photograph. Similarly, if the right slit is covered and the left slit open, the photograph will look like that in Figure 4.5. If both slits are open, Newton's particle picture of light leads to the prediction that the photographic plate will look like that in Figure 4.6, an amalgam of Figures 4.4 and 4.5. In essence, if you think of Newton's corpuscles of light as if they were little pellets you fire at the wall, the ones that get through will be concentrated in the two areas that line up with the two slits. The wave picture of light, on the contrary, leads to a very different prediction for what happens when both slits are open. Let's see this.

Imagine for a moment that rather than dealing with light waves we use water waves. The result we will find is the same, but water is easier to think about. When water waves strike the barrier, outgoing circular water waves emerge from each slit, much like those created by throwing a pebble into a pond, as illustrated in Figure 4.7. (It is simple to try this using a cardboard barrier with two slits in a pan of water.) As the waves emerging from each slit overlap with each other, something quite interesting happens. If two wave peaks overlap, the height of the water wave at that point increases: It's the sum of the heights of the two individual peaks. If two wave troughs overlap, the depth of the water depression at that point is similarly increased. And finally, if a wave peak emerging from one slit overlaps with a wave